



University of Essex

Online

Risk Identification and Digitalisation Strategy Report

Client: Pampered Pets

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Group: 3

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Table of Contents

1. Executive Summary	4
2. Current Business Risk Assessment (Status Quo)	4
2.1) Choice of methodology: ISO-IEC 27005	4
2.2) Current Threat and Risk Modelling.....	4
2.3) Current Network Architecture	6
2.4) Diagram 1: Current Network Architecture	7
3. Digitalisation Risk Assessment.....	8
3.1) Proposed Transformation Scope	8
3.2) Methodology Selection: STRIDE	8
3.3) Digitalisation Threat and Risk Modelling	8
3.4) Diagram 2: Proposed High Availability (HA) & Secure Access Architecture	10
3.5) Critical Analysis of proposed Digital Security Architecture	10
4. Strategic Business Recommendations	11
4.1) Response to Strategic Business	11
4.2) Final Verdict & Compromise Strategy	12
4.3) Implementation Timeline & Budget Allocation (Gantt Chart)	12
5. Conclusion	13
6. Reference	14
a) Appendix A: Current Business Risk Matrix	14
b) Appendix B: Digitalisation Risk Matrix	15

- **List of Abbreviations**

Abbreviation	Full Meaning
API	Application Programming Interface
DDoS	Distributed Denial of Service
EDR	Endpoint Detection and Response
ERP	Enterprise Resource Planning
GDPR	General Data Protection Regulation
HA	High Availability
ISO	International Organization for Standardization
IT	Information Technology
MFA	Multi-Factor Authentication
PCI-DSS	Payment Card Industry Data Security Standard
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
SIEM	Security Information and Event Management
SLA	Service Level Agreement
VAT	Value Added Tax
VLAN	Virtual Local Area Network
WAF	Web Application Firewall

1. Executive Summary

Pampered Pets is a successful bricks-and-mortar pet food retailer known for premium, locally sourced products. To facilitate business expansion, investor Orla O'dour has proposed funding a digital transformation. This report provides a comprehensive risk assessment of the company's current physical operations versus the proposed digitalisation strategy. It outlines specific cybersecurity threats, system availability gaps, and operational risks, providing actionable mitigation controls to ensure secure, sustainable business growth.

2. Current Business Risk Assessment (Status Quo)

2.1) Choice of methodology: ISO-IEC 27005

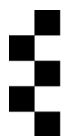
The current condition of Pampered Pets is assessed utilizing the ISO/IEC 27005:2018 framework.

Justification: ISO 27005 offers a systematic, asset-centric methodology for risk management. This approach is optimal for assessing physical vulnerabilities and quantifying risks in a conventional retail setting, as Pampered Pets is significantly dependent on tangible IT assets and human operating procedures (Sutton, 2021).

2.2) Current Threat and Risk Modelling

Currently, 90% of business is conducted face-to-face. While the local supply chain ensures exceptionally high product quality, the internal IT infrastructure is severely outdated and highly vulnerable to disruption.

Nº	Asset / Business Area	Threat	Vulnerability Analysis	Impact / Likelihood	Proposed Mitigations



Online

1	Network Infrastructure	Malware Infection & Data Sniffing.	The Front Desk PC (handling sales/tax) and Warehouse PC share a single Wireless Gateway with the personal smartphones of all four staff members.	High	Network Segmentation: Implement Virtual Local Area Networks (VLANs) to isolate business transaction systems from the staff's unmanaged personal mobile devices.
2	Data Integrity (Spreadsheets)	Accidental Data Deletion (Human Error).	Spreadsheets lack robust access controls. Critical VAT records can be accidentally overwritten.	High	Access Control: Enforce password protection on critical files and limit edit access based on the Principle of Least Privilege.
3	Warehouse User End-Points	Hardware Failure & Ransomware.	Harry uses an old, networked physical hardware device (PC) for inventory. This is a severe Single Point of Failure; if the hard drive fails, all inventory data is lost forever.	Medium	Endpoint Hardening: Upgrade endpoint hardware, strictly restrict unauthorized software installation, install Endpoint Detection and Response (EDR), and configure automated Cloud Backups.

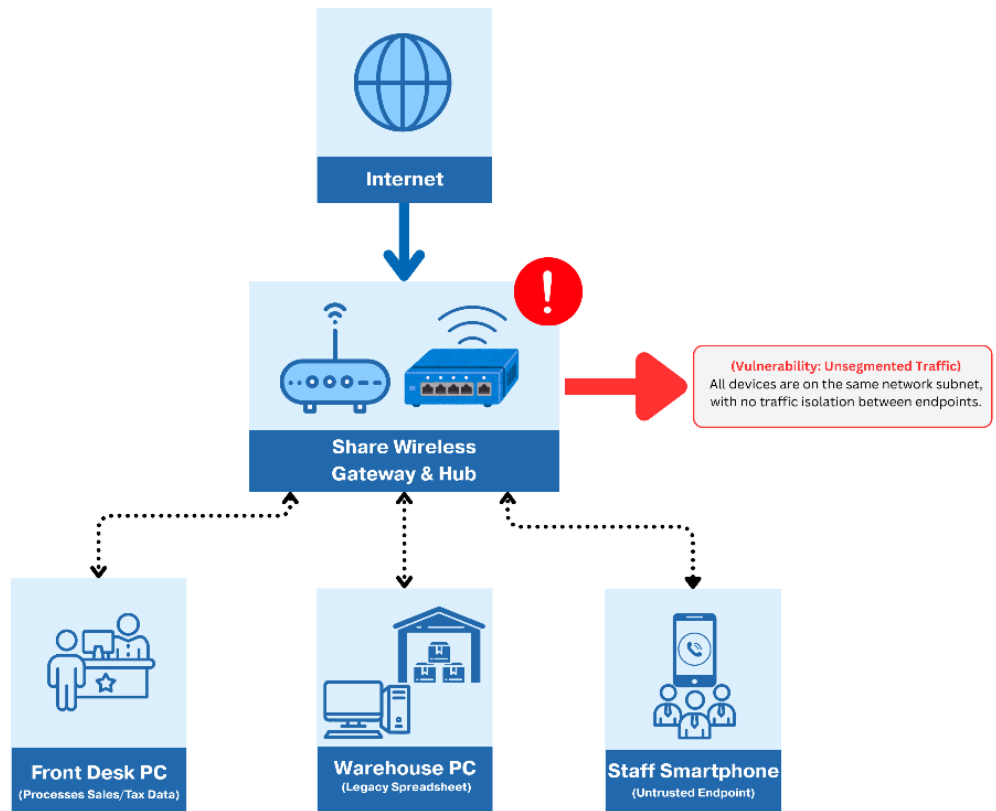


2.3) Current Network Architecture

Prior to threat modelling, key information assets were identified and rated by importance to the business:

Asset	Description	Importance
Customer Email Data	Order details, contact info - GDPR applicable	CRITICAL
Business Reputation	Brand trust built on quality products and local service	CRITICAL
Sales / Front Desk Computer	Processes all sales and purchases; VAT/tax records	HIGH
Warehouse Computer (networked)	Tracks deliveries, stock levels, item locations	HIGH
Wi-Fi Network & Hub	Connects all devices including staff smartphones	HIGH
Local Supply Chain	Farm suppliers within 10 minutes - quality guarantee	HIGH

2.4) Diagram 1: Current Network Architecture



The Core of Risks Detail:

- **Unrestricted Lateral Movement:** If malware infects a low-priority device (like a guest's smartphone or an outdated warehouse PC), the attacker can easily "move laterally" across the network. Because everything is connected, the infection can jump directly from that minor device to critical systems turning a minor breach into a total compromise of sensitive data.
- **Increased Attack Surface:** Every single device connected to the network acts as a potential unlocked door. An attacker only needs to compromise the weakest link to gain access to your highest-value targets.
- **Reduced Visibility and Control:** When all network traffic from vital databases to basic web browsing is intermingled in one open space, monitoring for active threats and controlling the flow of data becomes incredibly difficult.

3. Digitalisation Risk Assessment

3.1) Proposed Transformation Scope

To meet the goals of business growth and utilize Orla O'dour's funding, the following digital transformation is proposed:

- **High-Availability E-commerce Portal:** To capture national and global online sales.
- **Cloud ERP System:** To digitalise internal inventory and financial workflows.
- **International Supply Chain:** To reduce raw material procurement costs by 24%.

3.2) Methodology Selection: STRIDE

The STRIDE methodology is selected for the digital deployment phase.

Justification: STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) is the industry standard for analyzing software architectures, data flows, and web-facing cloud environments during system redesigns.

3.3) Digitalisation Threat and Risk Modelling

Migrating to an online model introduces complex threat surfaces. To close the security gap, enterprise-grade architecture controls must be woven directly into the implementation design:

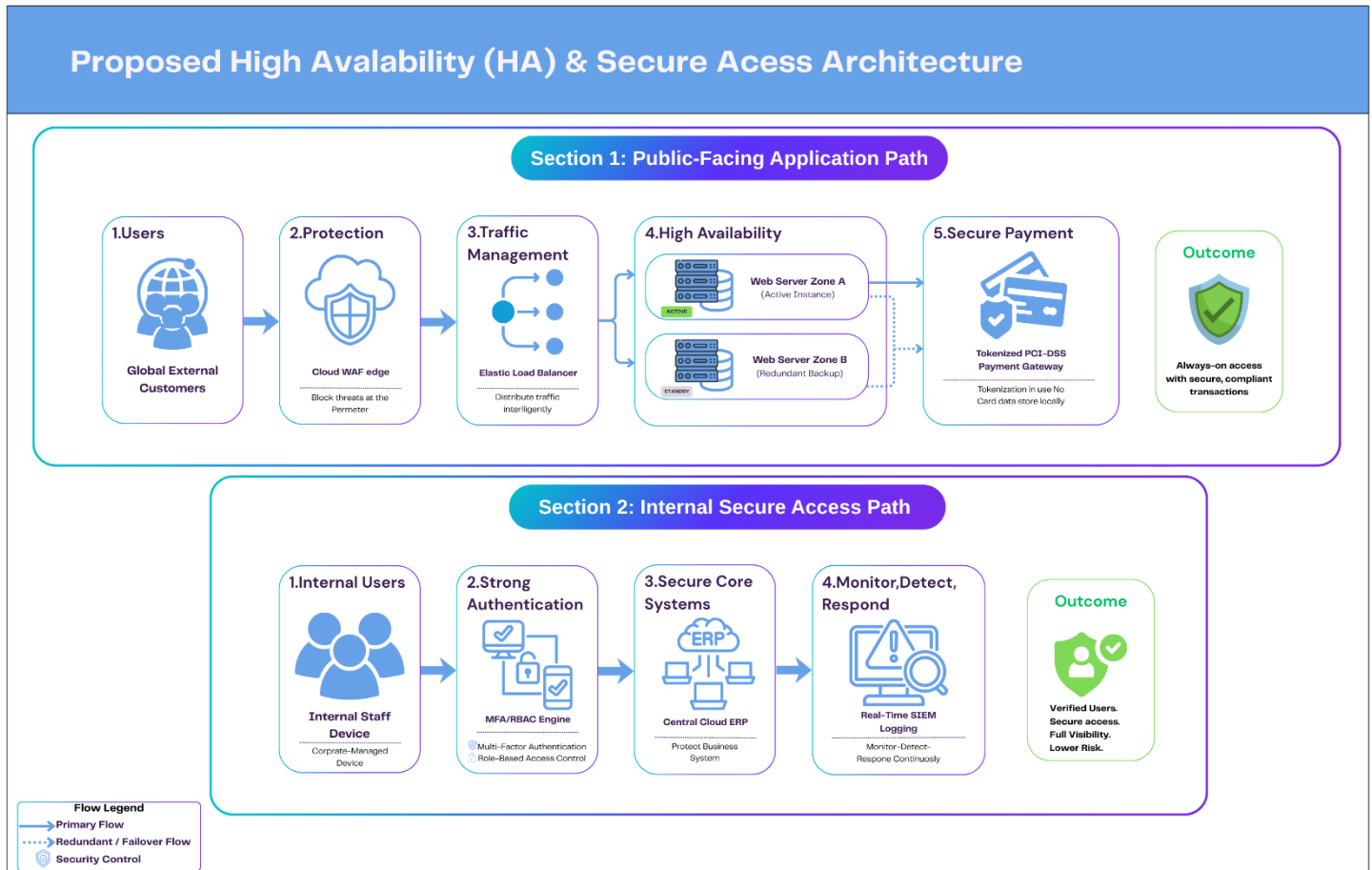
Nº	Proposed Change	Threat Analysis (STRIDE Category)	Impact / Likelihood	Proposed Mitigations
1	Int. Supply Chain	Operational Risk (Quality Drop): Losing the ability to physically check ingredient quality, risking the brand's premium reputation.	High	Vendor Risk Management: Implement strict Service Level Agreements (SLAs)



Online

				and require ISO 9001 quality certifications from all international suppliers.
2	Cloud ERP System	Elevation of Privilege: Attackers exploiting API vulnerabilities to compromise master inventory data.	High	SIEM Monitoring: Aggregate all identity logs into a Centralized SIEM platform. Mandate Multi-Factor Authentication (MFA) and Role-Based Access Control (RBAC).
3	Web Infrastructure	Denial of Service (DoS): Volumetric botnet attacks flooding web servers to cause platform downtime.	High	High Availability (HA): Route public traffic through an edge Cloud Web Application Firewall (WAF) and deploy a Load Balancer across multiple availability zones.
4	E-Commerce Portal	Information Disclosure: Web-facing threat actors exposing customer Personally Identifiable Information (PII).	Medium	Compliance Frameworks: Outsource payment processing to tokenized gateways to stay PCI-DSS compliant. Use end-to-end database encryption for GDPR alignment.

3.4) Diagram 2: Proposed High Availability (HA) & Secure Access Architecture



3.5) Critical Analysis of proposed Digital Security Architecture

- **Fault Tolerance via Load Balancing:** Splitting web infrastructure across independent physical Availability Zones protects the business from system downtime. If Server Zone A crashes due to a hardware failure, the Load Balancer automatically routes traffic to Zone B with zero user disruption.
- **Perimeter Edge Filtering:** The Cloud WAF blocks malicious traffic streams, SQL injections, and cross-site scripting (XSS) maneuvers before hackers can interact with core business databases.



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- **Offloaded Compliance Liability:** Tokenizing checkout processes via external payment gateways removes the risk of storing sensitive credit card records locally, protecting the brand from severe data breach liabilities.
- **Centralized Security Auditing:** Streaming system behaviors into a SIEM engine enables real-time automated alerting, allowing the technical team to flag and block anomalous login behavior immediately.

4. Strategic Business Recommendations

4.1) Response to Strategic Business

1) Could an online presence grow the business by up to 50%?

Yes. Introducing a high-availability E-commerce storefront bypasses the geographic limits of the retail shop, enabling national shipping and growing sales volumes beyond traditional walk-in limitations. Establishing an e-commerce presence can significantly expand the customer base beyond the local area (Whitman and Mattord, 2021).

2) Could changing to an international supply chain reduce costs by up to 24%?

Yes, but it introduces an unacceptable risk vector. Pampered Pets' core brand valuation rests on premium, locally sourced farm ingredients. Completely moving food operations abroad removes the 10-minute farm verification advantage, threatening the brand's core identity. International suppliers may offer lower unit costs; however, this must be weighed against longer lead times, reduced quality control, and loss of the guaranteed local supply chain (Calder and Watkins, 2019).

3) Could the business lose up to 33% of its existing customers if it doesn't provide online features?

Yes. Modern purchasing behaviors heavily favor businesses that offer digital features like click-and-collect or home delivery. Consumers increasingly expect businesses to offer at minimum a website or online ordering capability. Refusing to digitalise will cause immediate customer migration to tech-savvy competitors.



4.2) Final Verdict & Compromise Strategy

The consultancy team strongly recommends **proceeding with the digitalisation transformation** utilizing Orla O'dour's investment funding. The status quo risk of losing 33% of the established customer base is an existential danger to the company.

The "Hybrid Supply Chain" Compromise: To securely lower costs without degrading product value, the firm must implement a Hybrid Supply Chain. Pampered Pets will exploit international trade channels strictly for non-perishable hardware retail assets (e.g., toys, bowls, and leashes) to hit the targeted cost reductions, while firmly retaining local agricultural farms for their pet food raw materials to protect their premium quality.

4.3) Implementation Timeline & Budget Allocation (Gantt Chart)

- The project budget is systematically distributed across sequential operational phases to eliminate technical deployment gaps.

Nº	Project Phase & Milestone Targets	Timeline (Months 1-8)	Estimated Budget	Primary Funding Source
1	Phase 1: Core Perimeters & Infrastructure: - Deploy business VLAN segmentations - Perform endpoint hardening on physical hardware devices - Initialize automated cloud backup routines	1-2	\$5,000	Orla O'dour Capital Fund
2	Phase 2: HA Storefront Deployment: - Architect E-commerce portal across twin zones - Integrate tokenized PCI-DSS payment gateways - Position Cloud WAF edge policies	3-4	\$15,000	Orla O'dour Capital Fund
3	Phase 3: Automated Operations Sync: - Launch Cloud ERP inventory management engine - Implement centralized SIEM analytics infrastructure	5-8	\$25,000	Orla O'dour Capital Fund

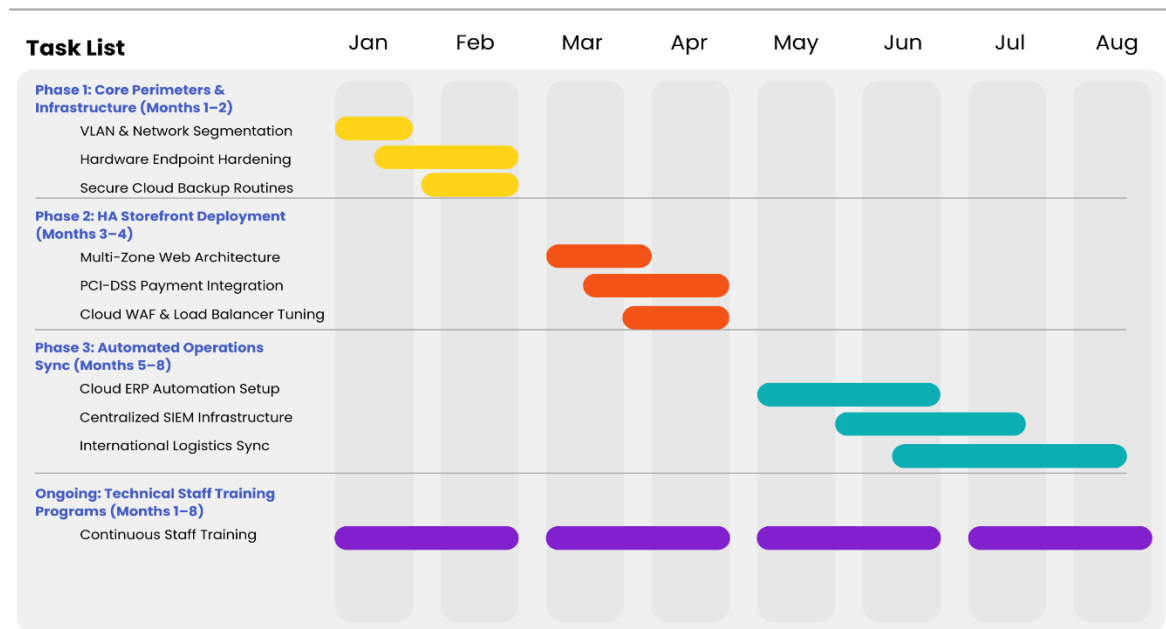
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	Onboard international hardware suppliers			
4	Ongoing: Technical Staff Training Programs Run continuous security awareness training.	1-8	\$5,000	Internal Operations Budget
TOTAL METRIC TARGETS		8 Months	\$50,000	100% Fully Funded

• Project Phase & Milestone Target:

Project Phase & Milestone Targets

The digitalisation transformation of Pampered Pets is structured into a strategic 8-month (32-week) phased timeline with a total estimated budget of \$50,000, fully funded by investor Orla O'dour. To eliminate technical deployment gaps and prevent any business downtime, the project is executed across distinct, sequential periods.



5. Conclusion

This report assessed the security and risk landscape for Pampered Pets across two scenarios: status quo and proposed digitalisation. The most pressing current threat is the potential loss of up to 33% of customers due to the absence of an online presence.

The proposed digitalisation introduces more complex risks around data protection and GDPR compliance, but these are mitigable. The STRIDE analysis confirms systematic threat planning is essential before go-live.

This team recommends a phased, security-first digitalisation. Both the 50% growth opportunity and the 33% customer loss risk strongly favour digital transformation. With Orla O'dour's investment, Pampered Pets has a credible path to sustainable growth.

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a) Appendix A: Current Business Risk Matrix

Risk matrix mapping Likelihood against Impact for the current business state.

Colour coding: RED = HIGH, AMBER = MEDIUM, GREEN = LOW.

Online

	Low Impact	Medium Impact	High Impact	Very High Impact
High Likelihood	Staff misuse of Wi-Fi	Unauthorised data access Wi-Fi interception	No online presence (33% loss risk) Malware/Ransomware	
Medium Likelihood	Physical theft	Customer data breach Supply chain disruption	Customer data loss	
Low Likelihood			Hardware theft	

b)Appendix B: Digitalisation Risk Matrix

Risk matrix mapping Likelihood against Impact for the proposed digitalisation scenario. Colour coding: DARK RED = CRITICAL, RED = HIGH, AMBER = MEDIUM, GREEN = LOW.

	Low Impact	Medium Impact	High Impact	Very High Impact
High Likelihood		Staff digital skills gap	Phishing attacks	
Medium Likelihood		Negative reviews Supply chain risks	Site hacked System failure	Payment data breach GDPR non-compliance
Low Likelihood			DDoS attack	